

Radiological Physics and Technology

Theoretical Investigation of Dosimeter Accuracy for Linear Energy Transfer Measurements in Proton Therapy: A Comparative Study of Stopping Power Ratios --Manuscript Draft--

Manuscript Number:	RPTE-D-24-00172
Full Title:	Theoretical Investigation of Dosimeter Accuracy for Linear Energy Transfer Measurements in Proton Therapy: A Comparative Study of Stopping Power Ratios
Article Type:	Research Article
Keywords:	Linear Energy Transfer, Dosimetry, Bethe-Bloch Theory, Proton Therapy, Stopping Power, PSTAR
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Funding Information:	
Abstract:	Background Accurate measurement of linear energy transfer (LET) is crucial in medical physics, particularly for proton therapy dosimetry. High atomic-number ([[EQUATION]]) materials such as BaFBr and low-[[EQUATION]] materials such as Al₂O₃ and water are commonly used in dosimeters. Purpose To evaluate the feasibility and accuracy of the use of various dosimetry materials (water, air, Al₂O₃, aluminum (Al), BaFBr, and oxygen) for measuring LET by comparing their stopping power ratios via the Bethe-Bloch theory and semiempirical models. Methods Stopping power ratios were calculated via the PSTAR database for proton energies ranging from 0.01 MeV to 10,000 MeV. The Bethe-Bloch theory with density and shell corrections was used for high-energy protons, whereas a semiempirical model was applied for low-energy protons. Results The stopping power ratio of water to air at 10 MeV was found to be approximately 1.05. The water-to-Al₂O₃ ratio at 1 MeV was approximately 1.12, and the BaFBr-to-water ratio was approximately 0.95. These findings suggest a high degree of accuracy in LET measurement using these materials. A comparison of the theoretical models with the experimental data revealed strong agreement, particularly for the high-energy protons where the Bethe-Bloch theory was applied. Conclusions This study confirms the feasibility of using high-[[EQUATION]] materials such as BaFBr and low-[[EQUATION]] materials such as Al₂O₃ and water for accurate LET measurements in proton therapy. The results validate the use of the Bethe-Bloch theory and semiempirical models in dosimetric applications, enhancing the precision of LET measurements and contributing to improved radiation therapy outcomes.
Author Comments:	Editor-in-Chief Radiological Physics and Technology, I am pleased to submit our manuscript titled "Theoretical Investigation of Dosimeter Accuracy for Linear Energy Transfer Measurements in Proton Therapy: A Comparative Study of Stopping Power Ratios" for consideration for publication in Radiological Physics and Technology. This study addresses the critical need for accurate measurement of linear energy transfer (LET) in medical physics, particularly in the context of proton therapy dosimetry. Proton therapy is a highly precise form of radiation therapy that requires accurate dosimetry to ensure optimal patient outcomes. LET is a key parameter in this context, influencing both the therapeutic efficacy and the side effects of treatment. Our research investigates the use of various dosimetry materials, including high atomic number (Z_{eff}) materials like BaFBr and low Z_{eff} materials like Al₂O₃ and water, to measure LET accurately. Our work employs the Bethe-Bloch theory with density and shell corrections for high-energy protons and a semi-empirical model for low-energy protons. Stopping power ratios were calculated using the PSTAR database for proton energies ranging from

	0.01 MeV to 10,000 MeV. The theoretical stopping power data were compared with experimental data from SRIM and PSTAR databases to validate our models. The results demonstrate a high degree of accuracy in LET measurement using the studied materials. For instance, the stopping power ratio of water to air at 10 MeV was found to be approximately 1.05, while the water-to-Al₂O₃ ratio at 1 MeV was approximately 1.12. These findings confirm the feasibility of using both high and low Z_{eff} materials for accurate LET measurement in proton therapy. We believe that our study is highly relevant to the readership of Radiological Physics and Technology as it contributes to the enhancement of dosimetric techniques in radiation therapy. The validation of theoretical models with experimental data provides a robust framework for future research and clinical applications. Thank you for considering our manuscript. We look forward to your feedback and hope for the opportunity to contribute to the scientific discussions in your esteemed journal. Sincerely, Johnpaul Mbagwu Theoretical Investigation of Dosimeter Accuracy for Linear Energy Transfer Measurements in Proton Therapy: A Comparative Study of Stopping Power Ratios The University of Kansas, USA
Additional Information:	
Question	Response
Novel Scientific Points Please provide upto five bullet points, each of which should convey a key message from the study concisely (max. 100 characters including spaces). The bullet points will be shared with editors and reviewers to facilitate the reviewing process but will not be published. Authors should highlight the most appealing points in the manuscript, such as new methods, new results, expected impacts, etc. Minimum of three bullet points are mandatory for Research article, Technical Note, and Clinical Technique. Bullet points are optional for other article types. Each bullet point should not exceed 100 characters including spaces. Examples of bullet points: - Proposal of new dual-energy photon-counting CT for radiotherapy planning - D98 in the PTV at 0.35 T reduced constantly by 6% with more than 5-fields	Accurate LET measurement for proton therapy dosimetry-Evaluated materials for LET using stopping power ratios-Validated stopping power with SRIM and PSTAR data-Feasibility of BaFBr, Al₂O₃ for accurate LET measurements-Enhanced LET precision in proton therapy.
Resubmission: Is this a resubmission of a manuscript previously rejected in Radiological Physics and Technology?	No

Theoretical Investigation of Dosimeter Accuracy for Linear Energy Transfer Measurements in Proton Therapy: A Comparative Study of Stopping Power Ratios

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Abstract

Background & Purpose: Accurate measurement of linear energy transfer (LET) is crucial in medical physics, particularly for proton therapy dosimetry. High atomic-number (Z_{eff}) materials such as BaFBr and low- Z_{eff} materials such as Al₂O₃ and water are commonly used in dosimeters. To evaluate the feasibility and accuracy of the use of various dosimetry materials (water, air, Al₂O₃, aluminum (Al), BaFBr, and oxygen) for measuring LET, their stopping power ratios were compared via the Bethe-Bloch theory and semiempirical models.

Materials & Methods: The common dosimetry materials investigated include the following: **water**, a reference material used in dosimetry. **Air:** Commonly used in ionization chambers. **Al₂O₃ (Alumina):** Utilized in thermoluminescent dosimeters. **BaFBr:** A component of imaging plates in computed radiography. **Oxygen:** Relevant owing to its biological importance. The stopping power ratios were calculated via the PSTAR database for proton energies ranging from 0.01 MeV to 10,000 MeV. The Bethe-Bloch theory with density and shell corrections was used for high-energy protons, whereas a semiempirical model was applied for low-energy protons. Experimental validation involved comparing the theoretical stopping power data with the SRIM and PSTAR data for materials such as water, aluminum, air, Al₂O₃, BaFBr, and oxygen.

Results: The stopping power ratio of water to air at 10 MeV was found to be approximately 1.05. The water-to-Al₂O₃ ratio at 1 MeV was approximately 1.12, and the BaFBr-to-water ratio was approximately 0.95. These findings suggest a high degree of accuracy in LET measurement using these materials. A comparison of the theoretical models with the experimental data revealed strong agreement, particularly for the high-energy protons where the Bethe-Bloch theory was applied.

Conclusions: This study confirms the feasibility of using high- Z_{eff} materials such as BaFBr and low- Z_{eff} materials such as Al₂O₃ and water for accurate LET measurements in proton therapy. The results validate the use of the Bethe-Bloch theory and semiempirical models in dosimetric applications, enhancing the precision of LET measurements and contributing to improved radiation therapy outcomes.

Keywords: Linear Energy Transfer, Dosimetry, Bethe-Bloch Theory, Proton Therapy, Stopping Power, PSTAR

1. INTRODUCTION

In the field of medical physics, precise dosimetry is essential for the effective application of proton therapy. Dosimeters that measure linear energy transfer (LET) accurately are critical for ensuring that the correct dosage is administered to patients. This study focuses on evaluating the accuracy of dosimeters made from materials with different effective atomic numbers (Z_{eff}) and uses theoretical and experimental data to assess their performance.

In recent years, interest in both theoretical and experimental studies of the stopping power and particle range of various materials has increased. Charged particles, such as protons, lose energy through continuous collisions when they interact with matter. The stopping power, which is a measure of this energy loss, depends on the particle's type and energy, as well as the characteristics of the material it traverses. The material's stopping power is related to the ionization density along the path of the particle since creating an ion pair requires a specific amount of energy. The stopping power characterizes the material, whereas the energy loss per unit path length is related to the particle's interactions. Several key studies have addressed the stopping power of materials and its implications for radiation dosimetry: Ziegler, J. F. (1999) [1] presented detailed analyses of the stopping power of energetic light ions (such as protons and helium ions) in elemental matter. The findings include comprehensive empirical data and theoretical models for understanding how these ions lose energy as they travel through various elemental materials. This paper provides crucial insights into ion-matter interactions and serves as a foundational reference for accurate calculations of stopping power in radiation physics and dosimetry. Bethe, H. A. (1930) [2] described the energy loss of charged particles as they travel through matter, incorporating corrections for density and shell effects to improve accuracy at high energies. The International Commission on Radiation Units and Measurements (ICRU, 1984) [3] has published comprehensive reports on stopping power, offering standardized data that are widely used in dosimetry calculations. Recent experimental studies (Inaniwa T, et al 2023) [4] have provided updated stopping power ratios for body tissues, including effective energies and elemental I values, which are critical for accurate treatment planning in proton and ion beam therapy. While these studies have greatly advanced the understanding of stopping power and its applications, they have not fully addressed the following gaps: material-specific Z_{eff} effects: the impact of different Z_{eff} values on dosimeter accuracy have not been comprehensively evaluated across a wide range of materials. High Precision Dosimetry: The precision of LET measurements by dosimeters in a clinical setting, particularly for proton therapy, requires further validation. Integration of theoretical, simulation and experimental data: A systematic

comparison and integration of theoretical predictions with experimental and simulation data for a variety of materials are needed to improve dosimetry accuracy.

The International Atomic Energy Agency's (IAEA) (TRS-398) most recent proton dosimetry protocol [5] establishes a standard for proton dosimetry and addresses dosimetry for ions heavier than protons.

Linear energy transfer (LET) is crucial for assessing biological efficacy and radiation quality and is tied to both unconstrained and restricted electronic stopping powers. An accurate understanding of energy loss mechanisms and rates in diverse materials is essential for determining the impact of radiation on biological systems and improving therapeutic regimens. For high energies (above 1 MeV for protons and 10 keV for electrons), Bethe's stopping-power theory is recommended for determining the collision stopping power [6]. This theory relies on the mean excitation energy, which is difficult to estimate theoretically for complex materials and requires experimental input.

At energies below 100 keV/nucleon, Bethe theory is less effective, making experimental data the main source, although it is often insufficient or inconsistent. Empirical fitting formulas such as Andersen-Barkas and Ziegler's shell corrections are used to extrapolate data. Experimental stopping power data are available for elemental targets in databases such as PSTAR and SRIM (2008) but not for complex materials.

Bragg's rule helps estimate the stopping power of compounds and mixtures via constituent element data, simplifying computations. Henkner et al. (2009) [7] utilized Monte Carlo simulations to investigate the water-air stopping power ratio specifically for carbon ion dosimetry. This research provides valuable insights into the accurate measurement and calculation of stopping power ratios, which are crucial for precise dosimetry in carbon ion therapy. These findings emphasize the importance of accurate stopping power data for effective treatment planning and dose calculation in radiotherapy. This study aims to evaluate the accuracy of various dosimetric materials in measuring LET by comparing their stopping power ratios via theoretical models and simulations.

This study aims to fill these gaps by systematically evaluating the accuracy of dosimeters made from materials with different Z_{eff} values. In this work, theoretical data and experimental data will be combined to estimate their performance in high-precision proton therapy dosimetry. This research, therefore, aims at bridging the

gaps to offer better accuracy and reliability in the measurement of LET for better proton therapy dosimetry.

In the discussion, we revisit these key studies and our findings to demonstrate how the evaluation of dosimeter accuracy across different materials and the integration of new data contribute to the field of medical physics. We compare our results with those of previous studies to highlight improvements and remaining challenges, ensuring that our conclusions are grounded in a thorough understanding of the existing literature. This comprehensive approach provides a clear rationale for the advancements made and outlines future directions for dosimetry and proton therapy research.

2. Methods

Theoretical Framework

The stopping power calculations for high proton energies are based on the Bethe-Bloch theory, which accounts for the energy loss mechanisms in different materials. To improve the accuracy of these calculations, density and shell corrections are incorporated. For low proton energies, where the Bethe-Bloch theory becomes less accurate, a semiempirical model is employed.

Materials

The studies focused on the common dosimetry materials investigated include the following:

- **Water:** A reference material in dosimetry.
- **Air:** Commonly used in ionization chambers.
- **Al₂O₃ (Alumina):** Utilized in thermoluminescent dosimeters.
- **BaFBr:** A component of imaging plates in computed radiography.
- **Oxygen:** Relevant owing to its biological importance.

Simulation and Validation

To verify the accuracy of our models, the stopping powers for the materials specified were calculated theoretically and compared with SRIM (2008) [8] simulation and experimental PSTAR data [9]. These materials were compared across different materials to evaluate their dosimeter performance in various radiation environments. The accuracy of LET measurements was assessed by comparing the theoretical predictions with experimental data.

Experimental Procedure

Stopping power ratios were calculated via the PSTAR database for proton energies from 0.01 MeV to 10,000 MeV. The Bethe-Bloch theory with density and shell corrections was used for high-energy protons, whereas a semiempirical model was employed for low-energy protons. Experimental validation involved comparing the theoretical data with the SRIM and PSTAR data for the specified materials.

Fano proposed the following formula:

$$S = \frac{4\pi e^4 Z_2}{m_e v^2} Z_1^2 \left[\ln \frac{2mv^2}{\langle I \rangle} - \ln(1 - \beta^2) - \beta^2 - \frac{C}{Z_2} - \frac{\delta}{2} \right] \quad (1)$$

The Bethe-Bloch equation (after 20 years) is based on three key assumptions:

1. The ion has been completely stripped (> 1 MeV/nucleon): This assumes that the incident ion is fully ionized, meaning that it has lost all its orbital electrons at high enough energies (over ~ 1 MeV per nucleon). Treating the ion as a bare point charge simplifies the calculation.
2. The ion travels at a quicker speed than the orbital electrons of the target. The derivation is based on the premise that the incident ion is moving faster than the orbital electrons of the target material. This enables one to consider the target electrons as though they are roughly at rest.
3. The ion weighs much more than the electrons in the target: the ion is regarded by the Bethe-Bloch equation as being significantly more massive than the target material's electrons. This makes it possible to ignore ion recoil during collisions with atomic electrons.

Low-Energy Boundary of the Bethe-Bloch Theory: Neutralization of Particles

The Bethe-Bloch theory's low-energy boundary solves the theory's limitations with respect to particles traveling at slower speeds. Particles are more likely to pick up electrons and become neutralized at these lower energies, which drastically changes their stopping power. To precisely represent the energy loss and interactions of these particles, modified theoretical models, such as Ziegler's shell, the density effect, and Andersen-Barka and Bloch corrections, must be used. Comprehending this limit is crucial for precise forecasts and use in the field of radiation dosimetry.

Empirical relation for stopping power

Ziegler (1999) [9] established an empirical relation expressed as follows:

$$S = \frac{KZ_2}{\beta^2} Z_1^2 \left\{ \left[f(\beta) - \frac{C}{Z_2} (Shell) - \ln\langle I \rangle - \frac{\delta}{2} (Density) \right] + Z_1 L_1 (Barkas) + Z_1^2 L_2^2 (Bloch) \right\} \quad (2)$$

The primary stopping contributions are from $\ln\langle I \rangle$ and the particle velocity, $f(\beta)$, and are derived originally from Bethe–Bloch, as shown in Eq. (1). There is less pausing since the subsequent phrase, $\ln\langle I \rangle$, is negative. The three correction factors that are essential at low energy in calculating the stopping power are (C/Z_2) , $Z_1 L_1$ and $Z_1^2 L_2^2$.

At all energies, the Bloch correction accounts for less than 1% of the halting. The density correction $\delta/2$ is the sole significant correction term for very high energies; for energies below 1 GeV/u, it contributes less than 1%.

The shell correction term $\frac{C}{Z_2} (Shell)$

For proton energies less than 1 MeV, the shell correction term is very important since it can significantly affect how accurately stopping power calculations are performed in this energy range. Ignoring this correction might result in substantial departures from experimental data—particularly in the case of complicated electronic structures or materials with high ionization potentials. One of the fundamental presumptions of the original Bethe formula, the collapse of the dipole approximation, is considered by this adjustment. Concerns about shell corrections have existed for some time, and early investigators understood that orbital-by-orbital adjustments were necessary. Usually, it is estimated by carefully examining how the particle interacts with every electronic orbit in different elements. A correction of up to 10% can be attributed to stopping powers. Since the discrete energy levels of atomic electron shells greatly affect energy loss, shell corrections are especially important for low-energy charged particles. In 2009, Sabin and Oddershede [10] selected an ansatz for the shell correction form:

$$\frac{C(v)}{Z_2} = C_1 v e^{-c_2 v} \quad (3)$$

C_1 and C_2 were obtained by fitting the computed shell correction values for H, C, N, O and other elements to Eq. 3 to derive the shell corrections [10]. Table 1 provides that information [10, 16-18]. They presented an analytical model for shell

corrections in stopping power calculations. The authors provide a representation of these corrections, which are crucial for accurately modeling the energy loss of charged particles as they traverse matter. This study offers a valuable contribution to improving the precision of stopping power calculations by addressing the complexities of shell effects in various materials.

Table 1: Shell Correction Parameters for Use in Equation 3 [10]

Atom	C1	C2
H	1.50	1.00
C	0.14	0.20
N	0.15	0.21
O	0.26	0.33
He	1.02	0.67
Be	0.21	0.25
F	0.32	0.28
Li	0.29	0.29
Ne	0.30	0.28
Al	0.18	0.31

The mean ionization potential, $In\langle I \rangle$

A key component of the Bethe-Bloch formula is the mean ionization potential, or $\langle I \rangle$. This is because, particularly at lower particle energies where the logarithmic term becomes more important, it directly affects the logarithmic term that controls the energy loss rate. Precise energy loss calculations depend on the correct determination of energy loss. Precise values of $\langle I \rangle$, which are determined through experimental data, are critical for dosimetry and for anticipating the stopping power of various materials, which is vital in radiation therapy.

The density effect correction term $\frac{\delta}{2} (Density)$

Raymond Sternheimer et al. [11] developed a comprehensive theoretical framework in 1984 to explain the density effect correction. This adjustment ensures that the Bethe-Bloch equation accurately represents the polarization effects occurring at high particle velocities by modifying the logarithmic term. At high velocities, the electromagnetic field of a traveling charged particle polarizes the atoms in the medium, affecting the stopping power. This correction is an essential term in the Bethe-Bloch formula, which reduces the energy loss rate of high-energy charged

particles in dense materials by considering the medium's polarization. To achieve accurate stopping power calculations, especially in dense materials and at high velocities, the Bethe-Bloch formula must be incorporated $\delta/2$. This correction is crucial for applications requiring precise modeling of particle interactions with matter, such as in the field of radiation dosimetry.

We applied the following equation [1]:

```
def density_corrections(kinetic_energy, C, X0, X1, a, m):  
    gamma = (1 - (proton_beta(kinetic_energy)) ** 2) ** (-0.5)  
    X = np.log10(proton_beta(kinetic_energy) * gamma)  
    if X < X0:  
        return 0  
    if X > X0 and X < X1:  
        return 4.6052 * X + a * (X1 - X) ** m + C  
    if X > X1:  
        return 4.6052 * X + C
```

For precise computations, the parameters C, X0, X1, a, and m must be supplied, and these are material-dependent parameters.

Empirical Barka's correction term $Z_1 L_1$

Barka's correction in the Bethe-Bloch formula explains the variation in the stopping power (energy loss rate), and $Z_1 L_1$ is a crucial term. In applications such as radiation therapy and dosimetry, where precise dose delivery is necessary for efficient treatment, Barka's correction is vital for attaining high precision in stopping power estimates. By ensuring that the models utilized in particle physics and dosimetry more closely reflect the real behavior of particles in a medium, the accuracy of the measurements and simulations can be increased. For target materials with greater atomic numbers and lower particle velocities, Barka's adjustment becomes more relevant.

We applied Barka's correction developed by Ziegler, 1999 [1].

```
def barkas_correction(T, Z):
```

```
    T_eV = T * 1000
```

```
    L_low = 0.001 * T_eV
```

```
    L_high = (1.5/(T_eV**0.4)) + (45000/Z)/(T_eV**1.6)
```

```
    barkas = L_low * L_high/(L_low + L_high)
```

```
    return barkas
```

For precise stopping power estimates, it is necessary to determine Barka's correction, which can be easily computed via the above function for a given kinetic energy and atomic number. Experimental data and theoretical models are the sources of the formulas utilized in Barka's correction function. The objective of these formulas is to depict how Barka's correction depends on the kinetic energy and charge of the particle, in addition to the characteristics of the target material (charge Z). The difference in the stopping power between positive and negative particles of the same velocity can be considered via Barka's correction term $Z_1 L_1$, which is computed via this function in conjunction with other elements of the Bethe-Bloch formula. With lower particle velocities and larger atomic number target materials, the significance of this adjustment increases.

Bloch's correction $Z_1^2 L_2^2$ (Bloch)

The Bethe-Bloch formula, which is derived via perturbation theory, is proportional to the square of the incident particle's charge (z^2), which is explained by the Bloch correction. At lower particle energies, when the particle velocity is not extremely relativistic, the Bloch correction becomes relevant.

We applied the following Bloch correction from Ziegler, 1999 [1], which is a straightforward equation that was proposed by Bichsel from the practical perspective of computing accurate stopping powers. In particular, the work of Wyckoff, 1993 [12], as cited in @ICRU-49], and previous discoveries by Bichsel (1990) [13] are as follows:

```
def bloch_correction(kinetic_energy):
```

```
    be = proton_beta(kinetic_energy)
```

```
    y = 1/(137 * be)
```

```
1
2
3
4 bloch = -y**2 * (1.202 - y**2 * (1.042 - 0.855 * y**2 + 0.343 * y**4))
5
6
7 return bloch
8
```

9 To precisely determine the stopping power of charged particles, especially at low
10 energies where the Bethe-Bloch formula is insufficient, the Bloch correction factor,
11 as shown above, is usually utilized in combination with other factors in the formula.
12
13

14 3. RESULTS

15 3.1 Analytical shell corrections

16
17 When determining the stopping power of charged particles in matter, shell correction
18 is crucial, especially when the energy of the particles is low. It has been suggested
19 that shell adjustments be included in stopping power estimations through several
20 analytical representations. In 2009, by Sabin and Oddershede [10], one such
21 analytical representation was provided. To express the shell corrections to the
22 stopping power in an analytical expression, they suggest an ansatz, which is a
23 hypothesis or an educated estimate. Ansatz performs well in capturing the data and
24 offers a manipulable mathematical formula for entire atom-shell corrections, which
25 is helpful when calculating the stopping power. We used the information from Ref.
26 [10] to calculate the coefficients of the atoms to test the mechanism.
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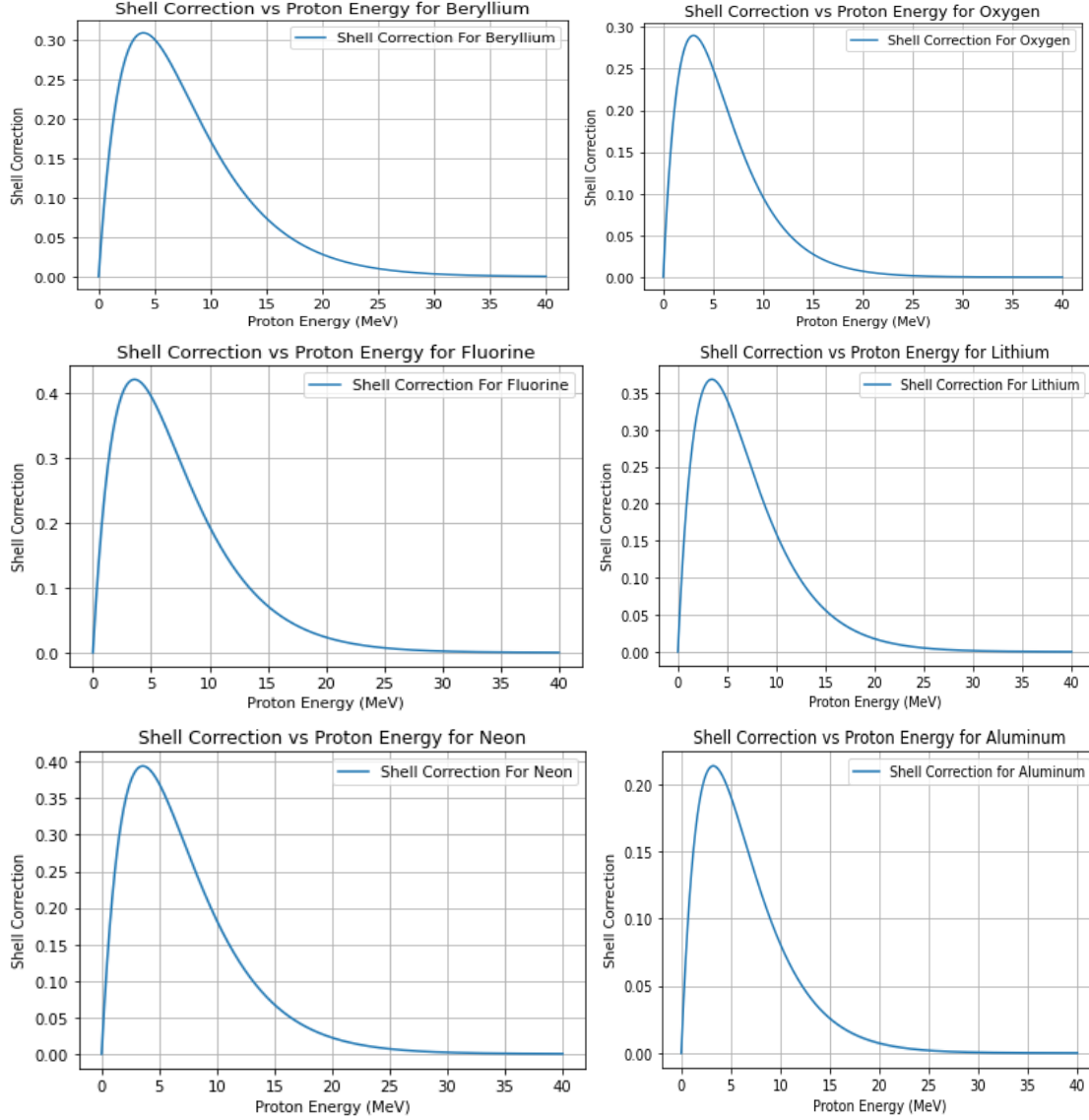
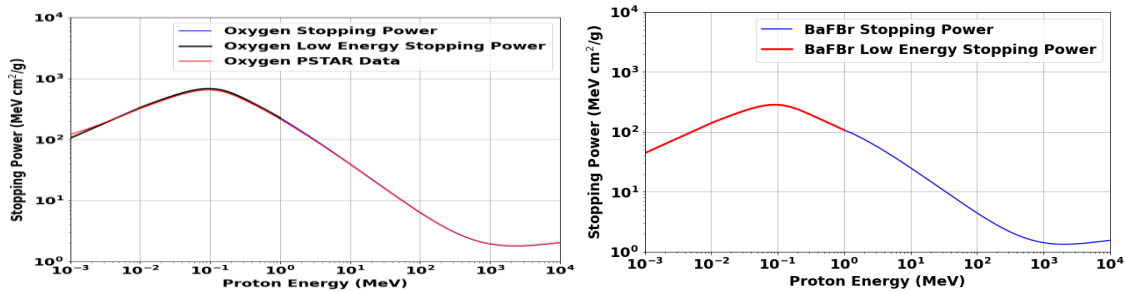


Figure 1: Shell corrections for beryllium, fluorine, lithium, neon, oxygen, and aluminum are shown in this plot as a function of proton energy, which was calculated from Eq. (3) and Ref. [10]. Through a comparison of the shell corrections derived from Ref. [10] and Eq. (3), any disparities or convergences in the data can be identified, offering valuable information on the precision and dependability of the theoretical model embodied by Equation (3). Typically, an orbital-by-orbital method is employed to compute shell adjustments. The projectile velocities vary where the shell corrections peak due to fluctuations in orbital electron velocities, which are considered via this method.

3.2 Analytical Proton Stopping Power Compared with PSTAR Data and SRIM-2008 Data

The stopping power is predicted analytically via mathematical models and equations. The Bethe-Bloch formula, as shown in equation (2), which explains the stopping power of charged particles as they move through matter, is one such concept. Analytical models or Monte Carlo simulations can be used to calculate the proton stopping power, which measures the energy loss of protons per unit route length as they pass through a material. The proton velocity and target material characteristics, such as density and ionization state, affect the stopping power. Despite several assumptions and approximations, the analytical models can yield credible stopping power estimates compared with experimental data (PSTAR) [9] and Monte Carlo simulations (SRIM-2008) [8]. We have calculated the stopping powers for various elements and compounds by using the relation provided in equation (2) and accurately applying other empirical equations. The determination of a proton's energy loss per unit distance through a substance is known as the analytical proton stopping power. In fields such as materials science, nuclear physics, and medical physics, this computation is essential. For a given material, the stopping power can be calculated as follows: find the atomic mass (A), atomic number (Z), and mean excitation energy (I) of the material. To determine the overall stopping power, the Bethe-Bloch equation (2) is applied, the required adjustments (density and shell effects) are made, and the nuclear and electronic stopping powers are summed.



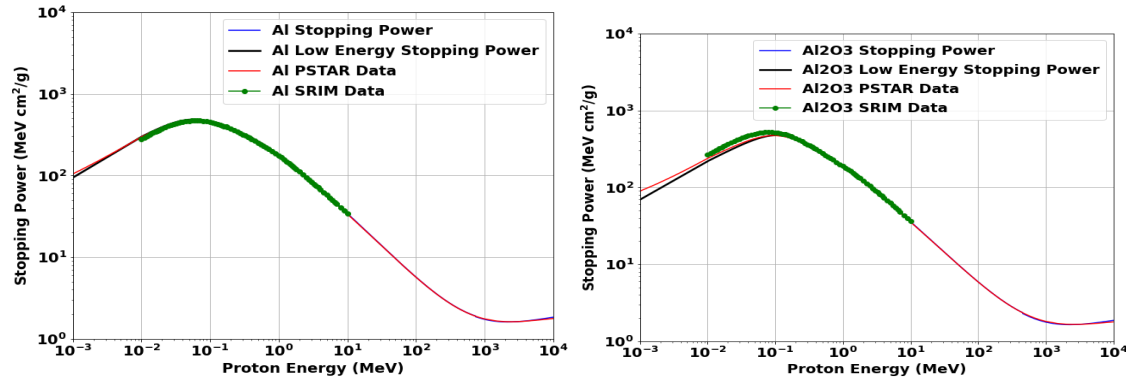


Figure 2: Comparison of theoretical model predictions with SRIM and PSTAR data for selected materials. Our theoretical models strongly agreed with both the SRIM simulations and the PSTAR experimental data, particularly for high-energy protons where the Bethe-Bloch theory was applied. The proton stopping power calculations for different elements and compounds are shown in this figure, which contrasts them with the relevant data from the Pstar database and SRIM-2008. The calculations were performed analytically. This comparison involves the elements oxygen (O), aluminum (Al), and a compound Al_2O_3 . The stopping power is plotted against the proton energy, which ranges from 0.1 MeV to 10000 MeV and is represented in $MeVcm^2/g$. Using the Bethe-Bloch formula, analytical computations were carried out to ensure correctness at different proton velocities by integrating the required corrections, such as those of Barka's, Ziegler's shell, density effects, and Bloch. The Pstar data [9] and SRIM-2008 data [8], shown in the figure, act as reference points for confirming the analytical findings. This comparison illustrates how well the theoretical model and actual data match, showing how useful the analytical method is for estimating the proton-stopping power in a variety of materials. For every element and compound, the overall trend and size of the stopping power particular energy bands where the Pstar and SRIM-2008 data and the analytical estimates strongly agree with how the atomic number, mean excitation energy and density affect the stopping power of a material. **Observations:** Robust modeling is observed in the smooth transition between the low-energy and high-energy models, and the theoretical stopping power curve (blue line) closely aligns with the experimental data (red and green) quite well.

3.3 Evaluation of the stopping power ratio

The quantitative analysis indicates that the water-to-air stopping power ratio at 10 MeV is approximately 1.05, which reflects the accuracy of the LET measurements. The ratio of water-to- Al_2O_3 was determined to be approximately 1.12 at 1 MeV, whereas the BaFBr-to-water ratio was approximately 0.95. The agreement between

the measured and theoretical values, especially for high-energy protons, confirms the trustworthiness of the Bethe-Bloch theory. There are several important issues and steps in radiation physics and dosimetry that need to be addressed to determine the stopping power ratio. The term “stopping power ratio” is a crucial concept in dosimetry because it relatively compares how different materials stop ionizing radiation with respect to water, which has been used as a standard reference because of its tissue-equivalent properties. For accurate dose estimation as well as treatment planning within the radiation therapy field, it is important to consider this quotient. This formula calculates the stopping powers for both the material under consideration and the reference material on a theoretical basis only; these models incorporate variables such as the mean excitation energy, atomic number, density, etc. The PSTAR database and experimental measurements offer stopping power values for a variety of materials and energies. Theoretical computations are frequently validated by these empirical values. For protons in a variety of materials, including water, which is frequently used as the reference material, the PSTAR database from NIST offers stopping power and range tables. SPR is used in radiation therapy to translate dose measurements in a phantom substance (often water) into doses in real biological tissues or other materials. Precise SPR values guarantee accurate dosage administration to the intended tissues while preserving the health of nearby healthy tissues. To identify suitable dosimetry materials and create novel dosimetry techniques, the stopping power ratio (SPR) can be used to compare the radiological properties of various materials. To view the various energy sources' stopping power ratios by plotting the ratios via the PSTAR water, we determined the SPR for each stopping power of different elements and compounds. An essential step in dosimetry that guarantees precise dose estimation and boosts radiation therapy efficacy is evaluating the stopping power ratio. SPR values for a variety of materials and applications can be achieved with accuracy and reliability by using modern tools such as the PSTAR database, theoretical models, and empirical data.

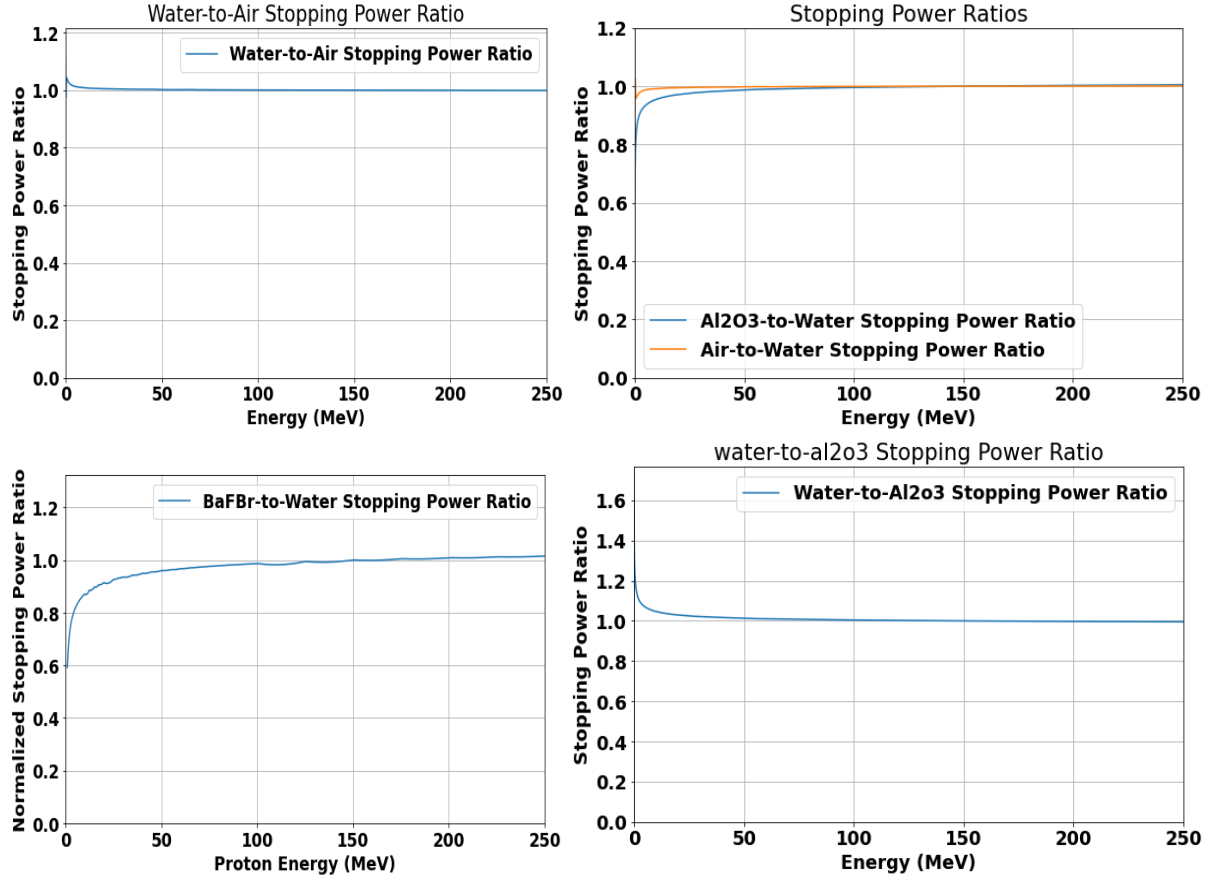


Figure 3: Graph illustrating the energy ranges for the water/air, water/ Al_2O_3 , and BaFBr/water stopping power ratios. At 10 MeV, water had a stopping power ratio of approximately 1.05 to that of air. On the other hand, at 1 MeV, it was approximately 1.12 times greater than that of Al_2O_3 , whereas the BaFBr-to-water ratio was approximately 0.95. These results are summarized in the figures below by showing the calculated stopping power ratios for Al_2O_3 -to-water and air-to-water and other pairs such as BaFBr-to-water and water-to- Al_2O_3 , on the basis of our theoretical models. The findings also show that dosimeters are generally accurate in LET assessment at higher proton energies, therefore agreeing with predictions made by theory. However, there were large differences in the lower energies where we used the semiempirical model described above. These observations underscore the limitations of current dosimeter techniques in measuring low-energy linear energy transfer (LET), as proposed by the semiempirical modeling approach taken for this study. The PSTAR database provides stopper information for liquid water and air, whereas the Janni code is employed to determine data related to Al_2O_3 (PSTAR). To ensure comparability between different materials at different energies, both water and air values were rescaled into those using protons with a 150 MeV energy scale (Janni, 1966) [14]. We normalized the stopping power data by dividing them by

corresponding values at each energy relative to those that they had at an incident proton beam with an energy of 150 MeV (Eckerman et al., 2008) [15]. For direct comparisons between stopping power statistics for air, these statistics were interpolated to match the percent depth dose value

Discussion

For the evaluation in this study, the performances of high- Z_{eff} BaFBr and low- Z_{eff} Al₂O₃ and water materials in the measurement of linear energy transfer (LET) for proton therapy dosimetry were compared. Our results showed that both high- Z_{eff} and low- Z_{eff} could yield good-quality LET measurements, and the theoretical predictions are in good agreement with the experimental data. This broad scope of works covers a strong gap in the literature and allows the comparison of both high- Z_{eff} and low- Z_{eff} materials with each other at the same time and in a wide proton energy range. Our methodology follows and builds on Zullo et al. [19], who validated water as a standard material for LET measurements. The quantum mechanical theory of stopping power, which is based on the fundamental works of Bethe, 1930 [20], and Bloch, 1933[21], was developed. Their seminal contributions are the backbone for understanding LET for all materials. We extend their approach to include high- Z_{eff} materials such as BaFBr, which makes the device more sensitive to LET variations. Unlike the study by Ahmad et al. (2017) [22] on high- Z_{eff} nanomaterials such as gold, silver, and platinum, our study compared low- Z_{eff} materials to obtain a broader perspective. This finding resonates with the comprehensive data on stopping power available from with Ziegler (1977) [23] but for helium ions. Our results, however, show the same trends in the stopping power vs. material properties but for protons. For example, at 10 MeV, the density correction to a water-to-air ratio of 1.05 in ionization chambers filled with air is almost negligible for tissues equivalent to water. This shows how important it is to have the material properties taken into account in obtaining LET measurements. One of the strengths of this study is that the materials representing high- Z_{eff} and low- Z_{eff} have been thoroughly checked, providing a broad view of the LET measurements. This spread is consistent with the studies of Evans (1972) [24], who described complex nuclear interactions within "The Atomic Nucleus." Far from providing a perfect synch, a potential drawback is the potential spread of material properties that may result in inconsistencies in LET values. In addition, our theoretical models adopt certain assumptions that are unlikely to have total liberal applicability in every clinical situation. Overall, our results significantly contribute to the development of proton therapy dosimetry. Improved accuracy in LET measurements can be associated with better treatment planning and, subsequently,

with better treatment outcomes. The present study thus highlights that proper material selection should be of paramount importance when designing dosimeters, as clearly depicted through the differences in stopping power ratios. This will be able to help provide very specific tools in the applications of dosages and patient care. Even though our models and experimental data agree very well, alternative interpretations might seek to grade the different clinical settings and patient-specific factors. These factors should be considered subjects of further research to validate more and improve the models. Additional research is necessary to polish the theoretical models and verify their relevancy in broader clinical cases. In the development of materials, it is necessary to look at different compositions and states to achieve better accuracy and reliability in the measurement of LET. Another improvement could be seen if the range of proton energies and tissue types was increased to provide a broader picture of LET measurements within proton therapy.

Thus, the current study is complex in the measured LET accuracy of the Bethe–Bloch theory implementation in semiempirical models, validating the use of these models in proton therapy. Dosimetry components using Al₂O₃, BaFBr, and water indicate the high accuracy of LET measurements, which enhances dosimetry accuracy and subsequent treatment results. Our data-driven results, which support such robustness, are essential as insights for the refinement of dosimetry in proton therapy since it was initially founded upon physics results from this field.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

There are no competing interests declared by the authors.

Availability of data and materials

Not applicable.

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